# Logistic Regression: From Intuition to Numerical Implementation

## Detailed Lecture Notes with Step-by-Step Examples

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**Abstract**

These comprehensive lecture notes cover Logistic Regression from intuitive understanding to detailed numerical implementation. This lecture notes provides clear explanations, step-by-step mathematical derivations, real-world applications, and practice problems with solutions. Designed for mathematics students transitioning to machine learning.

# 1 Introduction to Logistic Regression

**Key Insight**

Logistic Regression is NOT for predicting continuous values like Linear Regression. Instead, it's a **classification algorithm** that predicts probabilities between 0 and 1, making it perfect for "yes/no" decisions.

## 1.1 What Problem Does Logistic Regression Solve?

- **Binary Classification**: Predict one of two possible outcomes (Pass/Fail, Spam/Not Spam, Sick/Healthy)
- **Probabilistic Output**: Instead of just saying "yes" or "no," it says "70% chance of yes"
- **Linear Decision Boundary**: Finds the best straight line (or hyperplane) to separate classes

**Real-World Application**

**Common Applications:**

- **Medical Diagnosis**: Will this tumor be malignant (1) or benign (0)?
- **Credit Scoring**: Will this customer default on a loan (1) or not (0)?
- **Email Filtering**: Is this email spam (1) or not spam (0)?
- **Mobile Ads**: Will the user click on this ad (1) or ignore it (0)?

# 2 Why Linear Regression Fails for Classification

## 2.1 The Fundamental Problem

Consider our example dataset:

| Age (x) | Exam Result (y) | Interpretation |
|---|---|---|
| 20 | 0 | Failed |
| 30 | 0 | Failed |
| 40 | 1 | Passed |
| 50 | 1 | Passed |
| 60 | 1 | Passed |

**Mathematical Foundation**

**Linear Regression Attempt:**

$$y = mx + c$$

For classification, we want predictions between 0 and 1. But linear regression:

- Can predict values $< 0$ or $> 1$ (nonsensical for probabilities)
- Is sensitive to outliers
- Doesn't represent the S-shaped relationship we expect

## 2.2 The Categorical Data Challenge

- **Linear Regression**: Works with continuous target variables (like house prices, temperatures)
- **Logistic Regression**: Designed for categorical target variables (like pass/fail, yes/no)
- **Key Difference**: The output interpretation changes fundamentally

# 3 The Heart of Logistic Regression: Sigmoid Function

## 3.1 Mathematical Definition

**Mathematical Foundation**

**Sigmoid (Logistic) Function:**

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

**Where:**

- $z = b_0 + b_1x_1 + b_2x_2 + \cdots + b_nx_n$ (linear combination)
- $e \approx 2.71828$ (Euler's number)
- $\sigma(z) =$ probability that input belongs to class 1

## 3.2 Properties of Sigmoid Function

| Property | Explanation |
|---|---|
| **Range: (0,1)** | Output always between 0 and 1, perfect for probabilities |
| **S-shaped Curve** | Starts flat, rises steeply, then flattens again |
| **Decision Boundary at 0.5** | $\sigma(z) = 0.5$ when $z = 0$ |
| **Differentiable** | Smooth curve allows gradient descent optimization |
| **Monotonic** | As $z$ increases, $\sigma(z)$ always increases |

**Step-by-Step Example**

**Sigmoid Function Values:**

| $z$ | $\sigma(z)$ | Interpretation |
|---|---|---|
| $-5$ | 0.0067 | Virtually 0% probability |
| $-2$ | 0.1192 | About 12% probability |
| 0 | 0.5000 | Exactly 50% probability |
| 2 | 0.8808 | About 88% probability |
| 5 | 0.9933 | Virtually 100% probability |

Notice: Extreme z-values give probabilities near 0 or 1, while z=0 gives exactly 0.5.

# 4 Complete Numerical Example: Step-by-Step

## 4.1 Problem Setup

| **Age (x)** | **Result (y)** |
|---|---|
| 20 | 0 |
| 30 | 0 |
| 40 | 1 |
| 50 | 1 |
| 60 | 1 |

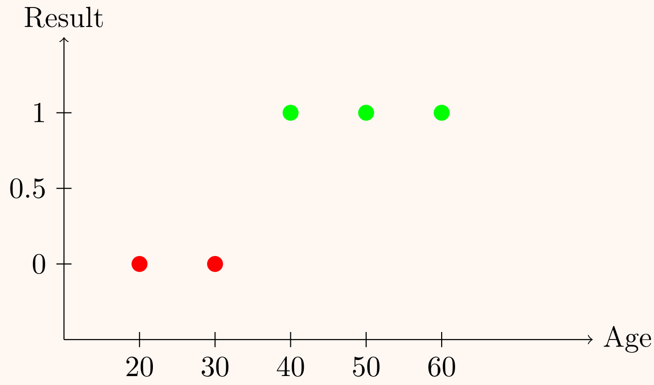
**Questions:**

1. Predict if a 15-year-old will pass the exam
2. Predict if a 45-year-old will pass the exam

## 4.2 Step 1: Visual Data Analysis

**Step-by-Step Example**

**Plotting the Data:**




Observations:

- Younger people (20,30) tend to fail
- Older people (40,50,60) tend to pass
- Clear pattern: More age = higher chance of passing

## 4.3 Step 2: Finding Linear Parameters

**Step-by-Step Example**

**Choosing Representative Points:** Let's pick two points that seem to represent the trend:

- Point A: (20, 0) - Young and failed
- Point B: (60, 1) - Old and passed

**Calculate Slope (m):**

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{1 - 0}{60 - 20} = \frac{1}{40} = 0.025$$

**Using Point-Slope Form:**

$$y - y_1 = m(x - x_1)$$

$$y - 0 = 0.025(x - 20)$$

$$y = 0.025x - 0.5$$

**Parameters for Logistic Regression:**

$$z = b_0 + b_1 x = -0.5 + 0.025x$$

Where: $b_0 = -0.5$ (intercept), $b_1 = 0.025$ (slope)

## 4.4 Step 3: Prediction for Age = 15

**Step-by-Step Example**

**Calculate z:**

$$z = b_0 + b_1 \times 15 = -0.5 + 0.025 \times 15$$

$$z = -0.5 + 0.375 = -0.125$$

**Apply Sigmoid Function:**

$$P(\text{Pass}) = \sigma(z) = \frac{1}{1+e^{-z}} = \frac{1}{1+e^{-(-0.125)}} = \frac{1}{1+e^{0.125}}$$

**Calculate e$^{\cdot 12}$:**

$$e^{0.125} = 2.71828^{0.125} \approx 1.1331$$

**Final Probability:**

$$P(\text{Pass}) = \frac{1}{1+1.1331} = \frac{1}{2.1331} \approx 0.468$$

**Classification Decision (Threshold = 0.5):**

$$0.468 < 0.5 \Rightarrow \text{Predict FAIL (Class 0)}$$

## 4.5 Step 4: Prediction for Age = 45

**Step-by-Step Example**

**Calculate z:**

$$z = b_0 + b_1 \times 45 = -0.5 + 0.025 \times 45$$

$$z = -0.5 + 1.125 = 0.625$$

**Apply Sigmoid Function:**

$$P(\text{Pass}) = \frac{1}{1+e^{-0.625}}$$

**Calculate e$^{\cdot 2}$:**

$$e^{-0.625} = \frac{1}{e^{0.625}} \approx \frac{1}{1.8682} \approx 0.535$$

**Final Probability:**

$$P(\text{Pass}) = \frac{1}{1+0.535} = \frac{1}{1.535} \approx 0.651$$

**Classification Decision:**

$$0.651 > 0.5 \Rightarrow \text{Predict PASS (Class 1)}$$

## 4.6 Step 5: Finding the Decision Boundary

**Step-by-Step Example**

**Decision Boundary Age:** The decision boundary occurs where $P(\text{Pass}) = 0.5$, which happens when $z = 0$:

$$0 = b_0 + b_1 x$$
$$0 = -0.5 + 0.025x$$
$$0.025x = 0.5$$
$$x = \frac{0.5}{0.025} = 20$$

**Interpretation:**

- Age $< 20$: More likely to fail ($P < 0.5$)
- Age $= 20$: Exactly 50% chance ($P = 0.5$)
- Age $> 20$: More likely to pass ($P > 0.5$)

This makes intuitive sense from our data pattern.

# 5 Mathematical Interpretation and Coefficients

## 5.1 Odds Ratio Interpretation

**Mathematical Foundation**

**From Probability to Odds:**

$$\text{Odds} = \frac{P}{1-P}$$

**Log-Odds (Logit Function):**

$$\log\left(\frac{P}{1-P}\right) = z = b_0 + b_1 x$$

**Odds Ratio:**

$$\frac{\text{Odds}(x+1)}{\text{Odds}(x)} = e^{b_1}$$

**Step-by-Step Example**

**Interpreting Our Coefficients:**

$$b_1 = 0.025 \Rightarrow e^{b_1} = e^{0.025} \approx 1.0253$$

Interpretation: For each additional year of age, the odds of passing increase by about 2.53%.
**Example Calculation:** For age 20: $z = -0.5 + 0.025 \times 20 = 0$

$$\text{Odds} = e^0 = 1 \Rightarrow P = \frac{1}{1+1} = 0.5$$

For age 21: $z = -0.5 + 0.025 \times 21 = 0.025$

$$\text{Odds} = e^{0.025} \approx 1.0253 \Rightarrow P \approx 0.506$$

## 5.2 Coefficient Significance

| Coefficient Value | Interpretation |
|---|---|
| $b_1 > 0$ | Positive relationship: as x increases, probability of y=1 increases |
| $b_1 < 0$ | Negative relationship: as x increases, probability of y=1 decreases |
| $b_1 = 0$ | No relationship: x doesn't affect probability of y=1 |
| $\|b_1\|$ large | Strong relationship: small changes in x cause large probability changes |
| $\|b_1\|$ small | Weak relationship: x has little effect on probability |

# 6 How Logistic Regression Learns from Data

## 6.1 Loss Function: Cross-Entropy

**Mathematical Foundation**

**Binary Cross-Entropy Loss:**

$$J(\theta) = -\frac{1}{n}\sum_{i=1}^{n}[y_i \log(\hat{y}_i) + (1 - y_i)\log(1 - \hat{y}_i)]$$

**Where:**

- $y_i$ = actual label (0 or 1)
- $\hat{y}_i$ = predicted probability
- $n$ = number of training examples

**Intuition:** Penalizes confident wrong predictions more than uncertain wrong predictions.

## 6.2 Gradient Descent for Optimization

**Step-by-Step Example**

**Gradient Descent Update Rule:**

$$b_0 := b_0 - \alpha\frac{1}{n}\sum_{i=1}^{n}(\hat{y}_i - y_i)$$

$$b_1 := b_1 - \alpha\frac{1}{n}\sum_{i=1}^{n}(\hat{y}_i - y_i)x_i$$

**Where:**

- $\alpha$ = learning rate (e.g., 0.01, 0.001)
- Iterate until convergence or maximum iterations

**Our Manual Approximation:** In our example, we manually estimated $b_0$ and $b_1$ from two representative points. In practice, gradient descent would find optimal values minimizing the loss function.

# 7 Practice Problems with Solutions

## 7.1 Problem 1: Basic Probability Calculation

**Practice Problem**

Given logistic regression parameters: $b_0 = -2$, $b_1 = 0.1$
For $x = 25$, calculate:

(a) The value of $z$

(b) The probability $P(y = 1|x)$

(c) The classification using threshold 0.5

**Solution:**

(a) $z = -2 + 0.1 \times 25 = -2 + 2.5 = 0.5$

(b) $P = \frac{1}{1+e^{-0.5}} = \frac{1}{1+0.6065} = \frac{1}{1.6065} \approx 0.622$

(c) Since $0.622 > 0.5$, classify as **Class 1**

## 7.2 Problem 2: Finding Decision Boundary

**Practice Problem**

A logistic regression model has parameters: $b_0 = -3$, $b_1 = 0.2$
Find the value of $x$ where $P(y = 1|x) = 0.5$

**Solution:**

$$0 = b_0 + b_1 x \Rightarrow 0 = -3 + 0.2x \Rightarrow 0.2x = 3 \Rightarrow x = 15$$

Decision boundary at $x = 15$

## 7.3 Problem 3: Multiple Predictions

**Practice Problem**

Given: $b_0 = -1$, $b_1 = 0.05$
Calculate probabilities for $x = 10, 30, 50$ and classify them with threshold 0.5

**Solution:**

$$x = 10: \quad z = -1 + 0.05 \times 10 = -0.5, \quad P = \frac{1}{1 + e^{0.5}} \approx 0.378 \Rightarrow \text{Class } 0$$
$$x = 30: \quad z = -1 + 0.05 \times 30 = 0.5, \quad P = \frac{1}{1 + e^{-0.5}} \approx 0.622 \Rightarrow \text{Class } 1$$
$$x = 50: \quad z = -1 + 0.05 \times 50 = 1.5, \quad P = \frac{1}{1 + e^{-1.5}} \approx 0.818 \Rightarrow \text{Class } 1$$

### 7.4 Problem 4: Real-World Scenario

**Practice Problem**

A bank uses logistic regression to predict loan default. The model is:

$$P(\text{Default}) = \frac{1}{1 + e^{-(-2.5+0.015\times\text{Income}-0.2\times\text{DebtRatio})}}$$

For a customer with Income = \$40,000 and DebtRatio = 0.3:

(a) Calculate probability of default

(b) If threshold = 0.3, will the bank approve the loan?

**Solution:**

(a) $z = -2.5 + 0.015 \times 40 - 0.2 \times 0.3 = -2.5 + 0.6 - 0.06 = -1.96$
$P(\text{Default}) = \frac{1}{1+e^{1.96}} = \frac{1}{1+7.10} \approx 0.123$

(b) Since $0.123 < 0.3$, predict "No Default" $\rightarrow$ **Approve Loan**

### 7.5 Problem 5: Model Interpretation

**Practice Problem**

In a medical diagnosis model: $b_0 = -3$, $b_{\text{age}} = 0.04$, $b_{\text{systolicBP}} = 0.02$
Interpret what happens when:

(a) Age increases by 10 years

(b) Systolic BP increases by 5 units

**Solution:**

(a) 10-year age increase: $z$ increases by $0.04 \times 10 = 0.4$
Odds ratio $= e^{0.4} \approx 1.492 \rightarrow$ odds increase by 49.2%

(b) 5-unit BP increase: $z$ increases by $0.02 \times 5 = 0.1$
Odds ratio $= e^{0.1} \approx 1.105 \rightarrow$ odds increase by 10.5%

## 8 Summary and Key Takeaways

### 8.1 Core Concepts Recap

**Key Insight**

**Fundamental Idea:** Logistic Regression takes linear regression and "squashes" it through a sigmoid function to get probabilities between 0 and 1.

| **Linear Regression** | **Logistic Regression** |
|---|---|
| Predicts continuous values | Predicts probabilities |
| Output range: $(-\infty, \infty)$ | Output range: $(0, 1)$ |
| Uses least squares loss | Uses cross-entropy loss |
| No activation function | Sigmoid activation function |

## 8.2 Step-by-Step Process

1. **Collect Data**: Get labeled training examples
2. **Define Model**: $P(y = 1|x) = \frac{1}{1+e^{-(b_0+b_1x)}}$
3. **Initialize Parameters**: Start with random $b_0$, $b_1$
4. **Compute Predictions**: Apply sigmoid to linear combination
5. **Calculate Loss**: Using cross-entropy formula
6. **Update Parameters**: Via gradient descent
7. **Repeat**: Until convergence
8. **Predict**: Apply trained model to new data

## 8.3 When to Use Logistic Regression

- **Good for**: Binary classification, probabilistic outputs, interpretable models
- **Not ideal for**: Highly non-linear relationships, very large feature spaces
- **Assumptions**: Linear relationship between features and log-odds
- **Data Requirements**: No multicollinearity, no outliers, large sample size

## 8.4 Common Pitfalls to Avoid

- **Perfect Separation**: When classes are completely separable, coefficients become infinite
- **Multicollinearity**: Correlated features cause unstable coefficient estimates
- **Small Sample Size**: Need sufficient data for reliable estimates
- **Non-linear Relationships**: May need feature transformation or polynomial terms

# Appendix: Useful Formulas Reference

**Mathematical Foundation**

**Key Formulas:**

$$\begin{aligned}
\text{Sigmoid:} \quad & \sigma(z) = \frac{1}{1+e^{-z}} \\
\text{Linear Combination:} \quad & z = b_0 + b_1x_1 + b_2x_2 + \cdots + b_nx_n \\
\text{Odds:} \quad & \frac{P}{1-P} \\
\text{Log-Odds (Logit):} \quad & \log\left(\frac{P}{1-P}\right) = z \\
\text{Odds Ratio:} \quad & e^{b_1} \\
\text{Decision Boundary:} \quad & z = 0 \Rightarrow b_0 + b_1x = 0 \\
\text{Cross-Entropy Loss:} \quad & J = -\frac{1}{n}\sum[y_i \log(\hat{y}_i) + (1-y_i)\log(1-\hat{y}_i)]
\end{aligned}$$

**Final Notes:** Logistic Regression remains one of the most widely used classification algorithms due to its simplicity, interpretability, and effectiveness. Understanding it thoroughly provides an excellent foundation for more complex machine learning models.

**Next Steps:**

1. Practice with real datasets (like Titanic survival prediction)
2. Learn about regularization to prevent overfitting
3. Explore multiclass extensions (Softmax Regression)
4. Compare with other classifiers (Decision Trees, SVM, Neural Networks)